

1. Personal Data

Date and place of birth	8 July 1991, Murska Sobota, Slovenia
Citizenship	Slovenian
Current residence	Kirkkonummi, Finland
Phone and email	+358 457 839 1902, jure.zrim@helsinki.fi

2. Education and degrees

2017–ongoing	DSc studies, Sustainable Use of Renewable Natural Resources, University of Helsinki. All required study credits completed; two of four dissertation articles published. Target graduation: 2027.
2022 – 2025	Career break for taking care of small children
2014 – 2016	MSc (Plant Production Sciences – Agroecology), University of Helsinki, October 2016.
2010 – 2013	BSc (Agronomy), University of Ljubljana, September 2013

3. Other education and training, qualifications and skills

Laboratory techniques: Soil DNA extraction, sterile laboratory practice, bacterial 16S and fungal ITS2 amplicon analysis, and earthworm identification.

Data analysis and programming: R, SPSS, Excel and Mothur sequence processing. Greenhouse gas flux and soil carbon stock calculations.

Environmental measurements: Static chambers, Gasmet DX 4015 FTIR and gas chromatography; soil C/N, bulk density, nutrient leaching, TDR, soil sampling, penetration resistance, aggregate stability, LAI and SPAD.

Driving licence: B

4. Linguistic skills

English	Full professional proficiency
Slovenian	Native proficiency
Finnish	Elementary proficiency

5. Present and previous professional appointments

2026	Researcher, Department of Agricultural Sciences, University of Helsinki.
2025	Visiting Researcher, University of Helsinki.
2021	Researcher, HYKERRYS 3, University of Helsinki; supervisor: Priit Tammeorg.
2019–2020	Researcher, HYKERRYS 2, University of Helsinki; supervisor: Priit Tammeorg.
2019	Researcher, Climsage, University of Helsinki; supervisor: Kristiina Karhu.
2018	Researcher, InCan: increasing soil carbon stocks and reducing nitrogen leaching with biochar; supervisor: Kristiina Karhu.
2017–	Doctoral researcher, University of Helsinki.
2016–2019	Researcher, HYKERRYS and TilaHYKERRYS, University of Helsinki; supervisor: Juha Helenius.
2015–	AgriChar research group: long-term effects of biochar and meat bone meal in Finnish agriculture; project leader: Priit Tammeorg.
2014	Research assistant in VORASUKA research project, University of Ljubljana

6. Research projects and other relevant experience

2026– NITROSCOPE: manage field experiments at two sites, coordinate greenhouse gas measurements and flux calculations, collect soil and crop data, and train student assistants. A 50% position is secured for January–December 2027; supervisor: Priit Tammeorg.

2025– PATH2CC: soil sampling, laboratory work, DNA extraction, microbial analyses and data quality control for soil carbon assessment. Contribute to carbon-credit assessment methods, monitoring workflows and soil health co-benefit evaluation; supervisor: Mari Sandell.

2026– CYANOACTION network: contribute to developing an online survey for stakeholders across the European Union.

2016–2022 HYKERRYS programme: trial planning, fieldwork and environmental measurements. Responsible for microbial analyses, interpretation and writing in the current recycled-fertilizer manuscript led by Mari Unnbom.

2019 COST Action ES1406 KEYSOM training school, Soil fauna – Key to Soil Organic Matter Dynamics and Modeling, University of Tartu, Estonia.

2019 Local arrangement team, 4th Finnish Biochar Seminar, Helsinki, 11 December.

2017 Local arrangement team, 22nd International Conference on Environmental Indicators, Helsinki, 1–5 August.

2016 Local arrangement team, 3rd Finnish Biochar Seminar, Helsinki, 25 November.

2015 Palopuro project, University of Helsinki: modelled biomass and nutrient recycling in an agroecological symbiosis; supervisor: Juha Helenius.

2014 VORASUKA, University of Ljubljana: field and laboratory experiments on rock-wool soil amendments in collaboration with Knauf Insulation; supervisor: Marina Pintar.

7. Research funding

2021 Kone Foundation: personal research grant, €30,000.

2019 Kone Foundation: personal research grant, €25,000.

2019 Finnish Cultural Foundation: personal research grant, €18,000.

2019 August Johannes and Aino Tiuran Foundation: personal research grant, €6,000.

2019–2020 Ministry of the Environment (Raki2), HYKERRYS 2: project funding €92,000. Project leader: Priit Tammeorg; role: soil microbial ecology and greenhouse gas research.

2018 August Johannes and Aino Tiuran Foundation: personal research grant, €6,000.

2018 Chancellor's Travel Grant, €2,000, for the 21st World Congress of Soil Science, Brazil.

2017–2021 Estonian Agricultural Registers and Information Board, Innovation Cluster of Field Crops: project funding €130,000. Project leader: Alar Astover; role: biochar researcher.

2017–2020 Uusimaa Centre for Economic Development, Transport and the Environment, TilaHYKERRYS: project funding €120,000. Project leader: Juha Helenius; role: biochar researcher.

2017 August Johannes and Aino Tiuran Foundation: personal research grant, €6,000.

2016 Olvi Foundation: personal research grant, €10,000.

2016–2019 Uusimaa Centre for Economic Development, Transport and the Environment, HYKERRYS: project funding €222,000. Project leader: Juha Helenius; role: part-time research assistant.

8**. Merits in teaching and pedagogical competence**

Supervision of theses **Supervised or co-supervised three MSc theses at the University of Helsinki:**

2026 (ongoing) Tabaraka Ali (co-supervisor, University of Helsinki)

2024 Zannatul Ferdous (co-supervisor, University of Helsinki)

2022 Vilja Jyrinki (supervisor, University of Helsinki)

2022 Eiko Höglund (supervisor, University of Helsinki)

9**. Other academic merits**

2017–2021 Member, Finnish Biochar Association

2018–2019 Board member, Finnish Biochar Association

2018 Vice President, Finnish Biochar Association

2017 Founding member, Finnish Biochar Association

2017– Personal research blog: jurezrim.wordpress.com

2016–2019 Member, Finnish Society of Soil Sciences

10**. Scientific and societal impact of research**

a) Publications

I have co-authored **three** peer-reviewed scientific journal articles and **twelve** non-refereed conference contributions (abstracts and posters), and authored **one** MSc thesis. Full references are listed below by publication type.

A1 Peer-reviewed scientific journal articles

1. Penttinen, P., Li, H., **Zrim, J.** & Lindström, K. 2026. Crop yield is associated with proportionality among soil microbiota. Geoderma, 475, 118092. https://doi.org/10.1016/j.geoderma.2026.118092
2. Kiani, M., **Zrim, J.**, Simojoki, A., Tammeorg, O., Penttinen, P., Markkanen, T. & Tammeorg, P. 2023. Recycling eutrophic lake sediments into grass production: A four-year field experiment on agronomical and environmental implications. Science of the Total Environment, 870, 161881. https://doi.org/10.1016/j.scitotenv.2023.161881
3. Kalu, S., Kulmala, L., **Zrim, J.**, Peltokangas, K., Tammeorg, P., Rasa, K., Kitzler, B., Pihlatie, M. & Karhu, K. 2022. Potential of biochar to reduce greenhouse gas emissions and increase nitrogen use efficiency in boreal arable soils in the long-term. Frontiers in Environmental Science, 10, 914766. https://doi.org/10.3389/fenvs.2022.914766

G2 MSc thesis

4. **Zrim, J.** 2016. Effect of softwood biochars on earthworms in two contrasting soils in Southern Finland. Master's Thesis. University of Helsinki.

B3 Non-refereed conference proceedings (including conference abstracts)

5. Roitto, M., Gao, Y., **Zrim, J.**, Jokiniemi, T. & Manca, M.-L. 2026. Creating value from carbon farming and co-benefits on small-scale farms. Poster presentation. 28th NJF Congress: Securing Food Supply in Nordic-Baltic Region for the Future, Tartu, Estonia, 14–16 October. https://www.emu.ee/en/njf2026
6. Unnbom, M., **Zrim, J.**, Helenius, J., **Tammeorg, P.** 2026. Recycled fertilizers and soil amendments in sustainable cropping systems – effects on soil carbon and microbial communities. Abstract of an oral presentation. Book of abstracts, p. 23–24. Nordic-Baltic scientific conference, 15–17 June 2026, Uppsala, Sweden.
7. Gao, Y., Roitto, M., **Zrim, J**., Jokiniemi, T., Tuomisto, H. & Sandell, M. 2026. PATH2CC Assessing carbon removals and co-benefits of carbon farming for small-scale farms. Poster. EU Carbon Farming Summit 2026, March 17–19.
8. Roitto, M. K., Gao, Y., **Zrim, J**., Jokiniemi, T., Tuomisto, H. & Sandell, M. 2026. PATH2CC – Hiilikrediittien mahdollisuudet pienille maatiloille Euroopassa. Poster. Maataloustieteen Päivät 2026.

B3 Non-refereed conference proceedings continued

9. Kiani M., Simojoki, A., Tammeorg, O., **Zrim, J**., Tammeorg, P. October 2021. Closing phosphorus cycle by reusing lake sediments in agriculture. Abstract of an oral presentation. Eurosoil 2021, Geneva, Switzerland.
10. Kiani, M., Simojoki, A., **Zrim, J.**, Tammeorg, O., Penttinen, P., Tammeorg, P. May 2019. Environmental Impacts of lake sediment reuse as a phosphorus-rich soil amendment. 15th International Conference of Sustainable Use and Management of Soil, Sediment and Water Resources, Antwerp, Belgium. Book of abstracts, p. 151.
11. **Zrim, J.**, Simojoki, A., Penttinen, P., Karhu, K., Tammeorg, P. 2019. How do softwood biochars affect the microbial communities and greenhouse gas emissions from agricultural soils in southern Finland? Poster abstract. X Maaperätieteiden päivät 2019, Helsinki, Finland, 9–10 January 2019. Pro Terra 75, p. 107–108.
12. **Zrim, J.,** Nuutinen, V., Simojoki, A., Penttinen, P., Glaser, B. & Tammeorg, P. 2019. Biochar impacts on earthworm and bacterial communities in two long-term field experiments in Finland. Poster abstract. Proceedings of the 21st World Congress of Soil Science, 2018, Rio de Janeiro, Brazil, August 12-17, Volume II, p. 422
13. **Zrim, J.**, Nuutinen, V., Simojoki, A., Penttinen, P., Karhu, K., Glaser, B. & Tammeorg, P. 2018. Effects of softwood biochars on soil biota in medium-term field experiments in Finland. Poster abstract. 3rd Conference on Ecology of Soil Microorganisms, 12–17 June, Helsinki, Finland. Book of abstracts, p. 209.
14. Tammeorg, P., **Zrim, J.**, Kiani, M., Mäkelä, P., Stoddard, F., Simojoki, A. 2017. Biochar as promising tool for carbon sequestration: Effects on Environment and Agriculture. Abstract of an oral presentation. The 22nd International Conference on Environmental Indicators, 1–5 August 2017, Helsinki, Finland. Book of abstracts, p. 22.
15. **Zrim, J.**, Nuutinen, V., Simojoki, A., Tammeorg, P. 2017. Effects of biochar on earthworms in two long-term field experiments in Finland. Poster abstract. The 22nd International Conference on Environmental Indicators, 1–5 August 2017, Helsinki, Finland. Book of abstracts, p. 48.
16. **Zrim, J.**, Nuutinen, V., Simojoki, A., Tammeorg, P. 2017. Effects of biochar on earthworms in two long-term field experiments in Finland. Poster abstract. IX Maaperätieteiden päivät 2017, Helsinki, Finland, 9–10 January 2017. Pro Terra 71, p. 99–100.